EXPENSE TRACKER CLI

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import csv
import os
from datetime import datetime

FILE_NAME = "expenses.csv"


def create_file():
    if not os.path.exists(FILE_NAME):
        with open(FILE_NAME, "w", newline="") as file:
            writer = csv.writer(file)
            writer.writerow(["Date", "Type", "Category", "Amount"])


def add_entry():
    print("\n--- Add Income / Expense ---")
    date = input("Enter date (YYYY-MM-DD): ")

    try:
        datetime.strptime(date, "%Y-%m-%d")
    except ValueError:
        print("Invalid date!")
        return

    entry_type = input("Enter type (Income/Expense): ").capitalize()

    if entry_type not in ["Income", "Expense"]:
        print("Please enter Income or Expense.")
        return

    category = input("Enter category: ")

    try:
        amount = float(input("Enter amount: "))
    except ValueError:
        print("Please enter a valid amount.")
        return

    with open(FILE_NAME, "a", newline="") as file:
        writer = csv.writer(file)
        writer.writerow([date, entry_type, category, amount])

    print("Entry added successfully!")


def view_entries():
    print("\n--- All Entries ---")

    with open(FILE_NAME, "r") as file:
        reader = csv.DictReader(file)
        found = False

        for row in reader:
            found = True
            print(row["Date"], "|", row["Type"], "|",
                  row["Category"], "| ■", row["Amount"])

        if not found:
            print("No entries found.")


def monthly_summary():
    print("\n--- Monthly Summary ---")
    month = input("Enter month (YYYY-MM): ")

    income = 0
    expense = 0

    with open(FILE_NAME, "r") as file:
        reader = csv.DictReader(file)

        for row in reader:
            if row["Date"].startswith(month):
                amount = float(row["Amount"])

                if row["Type"] == "Income":
                    income += amount
                elif row["Type"] == "Expense":
                    expense += amount

    balance = income - expense

    print("\nMonth:", month)
    print("Total Income : ■", income)
    print("Total Expense: ■", expense)
    print("Balance      : ■", balance)


def export_csv():
    output_file = "expense_report.csv"
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    with open(FILE_NAME, "r") as source:
        data = source.read()

    with open(output_file, "w") as destination:
        destination.write(data)

    print("CSV report created:", output_file)


def export_excel():
    try:
        from openpyxl import Workbook

        workbook = Workbook()
        sheet = workbook.active
        sheet.title = "Expense Report"

        with open(FILE_NAME, "r") as file:
            reader = csv.reader(file)

            for row in reader:
                sheet.append(row)

        workbook.save("expense_report.xlsx")
        print("Excel report created: expense_report.xlsx")

    except ImportError:
        print("Excel export requires openpyxl.")
        print("Install it using:")
        print("pip install openpyxl")


def create_chart():
    try:
        import matplotlib.pyplot as plt

        income = 0
        expense = 0

        with open(FILE_NAME, "r") as file:
            reader = csv.DictReader(file)

            for row in reader:
                amount = float(row["Amount"])

                if row["Type"] == "Income":
                    income += amount
                elif row["Type"] == "Expense":
                    expense += amount

        types = ["Income", "Expense"]
        amounts = [income, expense]

        plt.bar(types, amounts)
        plt.title("Income vs Expense")
        plt.xlabel("Type")
        plt.ylabel("Amount")
        plt.savefig("expense_chart.png")
        plt.show()

        print("Chart saved as expense_chart.png")

    except ImportError:
        print("Matplotlib is not installed.")
        print("Install it using:")
        print("pip install matplotlib")


def main():
    create_file()

    while True:
        print("\n================================")
        print("       EXPENSE TRACKER CLI")
        print("================================")

        print("1. Add Income / Expense")
        print("2. View All Entries")
        print("3. Monthly Summary")
        print("4. Export to CSV")
        print("5. Export to Excel")
        print("6. Create Chart")
        print("7. Exit")

        choice = input("\nEnter your choice: ")

        if choice == "1":
            add_entry()
        elif choice == "2":
            view_entries()
        elif choice == "3":
            monthly_summary()
        elif choice == "4":
            export_csv()
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        elif choice == "5":
            export_excel()
        elif choice == "6":
            create_chart()
        elif choice == "7":
            print("Thank you for using Expense Tracker!")
            break
        else:
            print("Invalid choice!")


main()
```